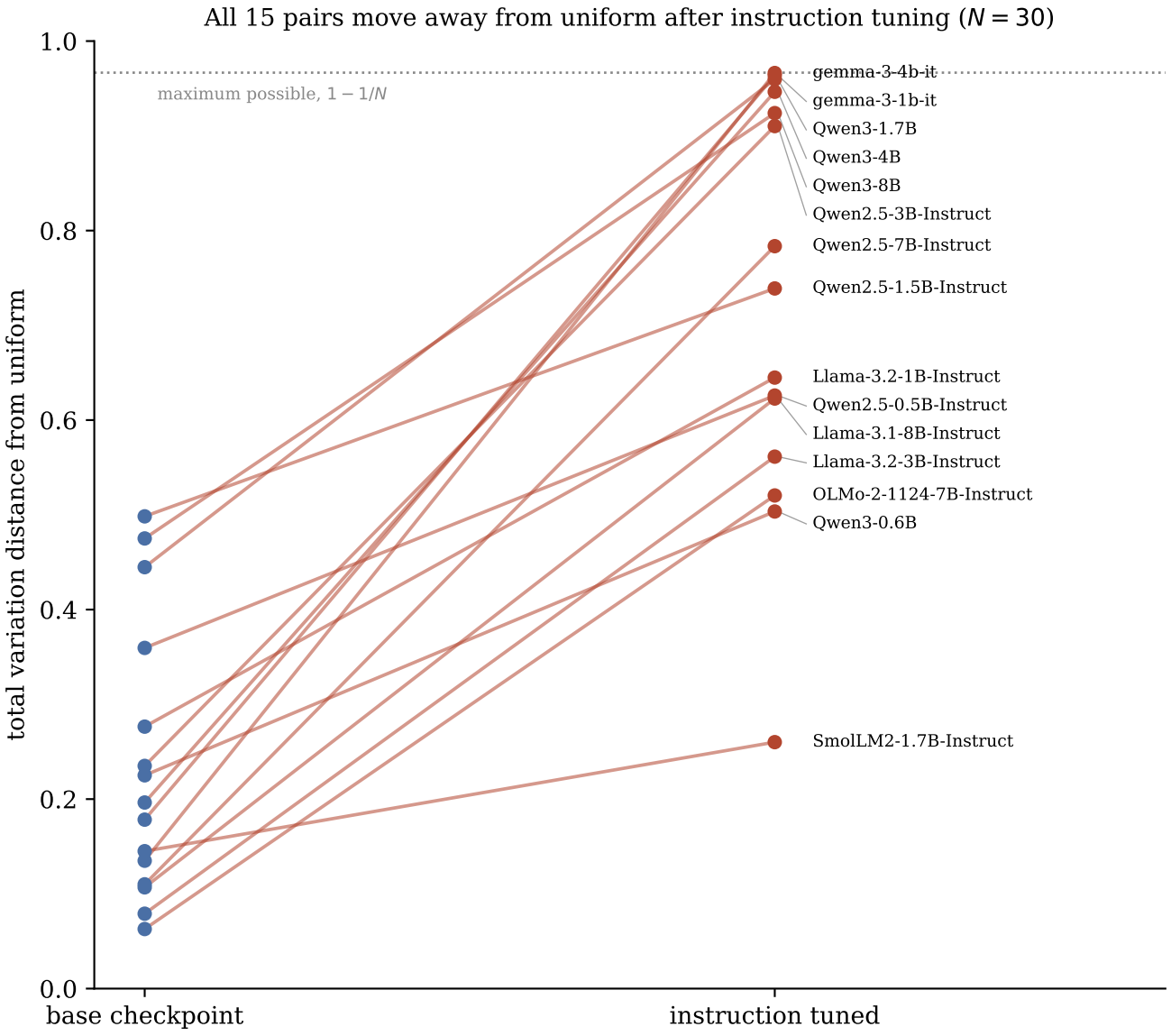
All 15 pairs move away from uniform after instruction tuning ($N = 30$)
total variation distance from uniform
1.0
0.8
0.6
0.4
0.2
0.0
maximum possible, $1 - 1/N$
base checkpoint
instruction tuned
gemma-3-4b-it
gemma-3-1b-it
Qwen3-1.7B
Qwen3-4B
Qwen3-8B
Qwen2.5-3B-Instruct
Qwen2.5-7B-Instruct
Qwen2.5-1.5B-Instruct
Llama-3.2-1B-Instruct
Qwen2.5-0.5B-Instruct
Llama-3.1-8B-Instruct
Llama-3.2-3B-Instruct
OLMo-2-1124-7B-Instruct
Qwen3-0.6B
SmolLM2-1.7B-Instruct